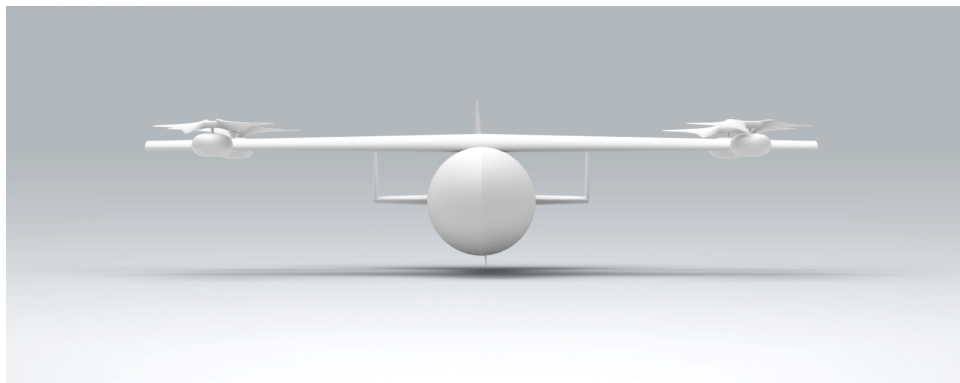




Virtual Flight Testing of a Small UAV for Aid Delivery at Sea Flying Life Raft



Project Coordinator: Dr Errikos Levis
Academic Supervisor: Dr Errikos Levis
Department: Department of Aeronautics
Academic year: 2021/2022

Module: AERO60004 Aeronautics Group Design Project
Subteam: Simulation & Testing
Student: Rohit Singh
CID: 01708106

Date: 20/06/2022

Abstract

Unmanned aerial vehicles (UAVs) open up promising capabilities in the field of search and rescue, with the ability for autonomous operations in hazardous environments. The Mediterranean Sea represents one such area, with over 3000 migrants lost at sea in 2021 alone. The flight testing of *Fregata*, a small hybrid-configuration UAV for aid delivery at sea, is presented. X-Plane 10 by Laminar Research is used to investigate its dynamic behaviour and the performance of the fixed-wing controller, and these results are compared to predictions. The dynamics behaviour in general differed from predictions due to inaccuracies in the neutral point prediction and aerodynamic deficiencies. The closed-loop performance of the aircraft is promising, and a limit-cycle oscillation in the roll controller was able to be suppressed. Future iterations of the aircraft should include a larger tailplane to increase the static margin along with smaller elevators to diminish the strong pitch authority.





FLR22	
Subteam	Members
Technical Directors	Foster Ng
	Julian Fung
	Daniel Earl
Aerodynamics	Anthony Haiser
	Kwok Leung Yim
	Marius Borsu Lefevre
Structures	Sorawis Itthithavorn
	Antoine Rieuneau
	Olivier Sussfeld
Propulsion and Systems	Jordi Frittoli
	Matan Hamburger
	Weihan Lee
	Natalia Melo Fernandez
	Farsai Sittichoksuwan
Performance, Stability & Control	Kristoff Ahlner
	Joshua Binns
	Elia Chimonides
	Luis Ferreira da Silva Marques
	Yi En Puah
Simulation	Bojun Vito Yang
	Gabisanth Seyon
	Rohit Singh



Contents

Team Members	i
Table of Contents	ii
List of Figures	iii
List of Tables	iv
1 Context and Motivation	1
2 X-Plane 10	1
2.1 Overview	1
2.2 Flight Physics	2
2.3 Plane Maker and Airfoil Maker	3
3 Modelling and Implementation	3
3.1 Modelling in Plane Maker	3
3.2 Communicating with X-Plane	4
3.3 Simulink Implementation	5
4 Flight Testing Scope	5
5 Flight Testing Results and Discussion	6
5.1 Initial Impressions	6
5.2 Phugoid Test	7
5.3 SPPO Test	8
5.4 Dutch Roll	9
5.5 Fixed-Wing Controller Performance	10
6 Conclusions	12
A Retuning Process for Pitch Controller	15

List of Figures

1.1	Proposed mission profile.	1
1.2	Render of baseline configuration.	1
2.1	Presented data of the propwash velocity distribution from YouTube [11], with induced velocity in blue and theoretical distribution in red.	2
2.2	Entire parameterised lift, drag, and moment curve for NACA 64 ₂ – 015 between -180° and 180° angle of attack.	3
3.1	Plane maker parameters for main wing, with discretised twist implemented.	4
3.2	Plane Maker model of <i>Fregata</i> compared to working CAD model.	4
3.3	Complete Simulink model for closed loop testing with key block systems highlighted. . . .	5
5.1	Sample control inputs for open loop system, with pilot-induced oscillations between 9 and 12 s.	7
5.2	Virtual trim system used for testing, with trim sliders highlighted.	7
5.3	Sample phugoid plot showing slight instability after 80 s.	7
5.4	Failed phugoid motion with stall following one oscillation.	8
5.5	Open-loop and closed-loop phugoid behaviour, with successful pitch control evident. . . .	8
5.6	Sample SPPO plot with oscillatory fit results.	9
5.7	Open-loop and closed-loop SPPO behaviour showing little difference between the two oscillations.	9
5.8	Sample Dutch roll plot with fit results.	10
5.9	Open-loop and Closed-loop Dutch roll behaviour showing successful damping of Dutch roll. .	10
5.10	Tracking for final altitude controller with improved performance.	11
5.11	Controller performance at 70 kn, with velocity deviation shown with stall speed.	11
5.12	Unstable roll controller response to reference signal, with limit cycle phase portrait. . . .	11
5.13	Successful roll control with unstable behaviour eliminated.	12
A.1	Retuning process for pitch controller, with final controller at bottom right.	15

List of Tables

1.1	Key parameters for <i>Fregata</i>	1
5.1	Characteristics of predicted and simulation dynamic modes.	8
5.2	Controller gains before and after simulator testing.	10

1 Context and Motivation

Unmanned aerial vehicles (UAVs) have revolutionised search and rescue capabilities, allowing for short response times to accidents and performing complex missions in increasingly hazardous environments [1]. The lack of a pilot and associated support systems means they can be very efficient, allowing for high endurance missions within a large search area. In particular, developments in control design have meant that the quadcopter configuration has become more and more feasible [2], allowing for vertical take-off and landing (VTOL) capabilities and excellent controllability. A desire for good range and low energy consumption gives rise to the hybrid fixed-wing-VTOL configuration, with the excellent cruise performance of a fixed-wing aircraft and the practicality of a VTOL vehicle.

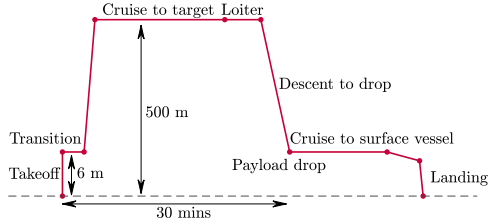


Figure 1.1: Proposed mission profile.

Fregata, a small VTOL capable UAV, has been designed by a group of students at Imperial College London to augment coast guard operations and deliver flotation aid to migrants involved in sea accidents in the Mediterranean Sea. The proposed mission profile and a render of the final configuration can be found in Figures 1.1 and 1.2, and a collection of important sizing parameters can be found in Table 1.1.

As part of the design process, it was desired that the vehicle be tested in a flight simulation software

to provide accurate predictions into its performance and dynamic behaviour. This report presents the virtual flight testing of the designed vehicle and performance investigations of the fixed-wing controller. For more information about the baseline configuration and design process, the reader is directed to the project executive summary [3], while for performance testing and VTOL and transition accuracy, the complementary report by Seyon [4] should be considered.

2 X-Plane 10

2.1 Overview

It was desired that a high-fidelity flight simulation software be used to carry out the flight testing. The simulation environment must capture more physics than the panel codes and state space methods used for aerodynamic and dynamic design, such as viscous effects and non-linearities in the system. X-Plane 10 is a professional flight simulation program from Laminar Research [5], initially developed by Austin Meyer [6]. The accuracy and versatility of the software is highly desirable and it was a clear contender for the simulation software for this project. An initial literature review revealed that at its core, X-Plane uses blade element theory for its flight physics model, brief details of which can be found below. It was found that this modelling approach has been shown to be a suitable framework for the simulation of both rotary wing [7] and fixed wing UAVs [8], and so X-Plane was initially deemed to be a suitable software for the simulation of our UAV.

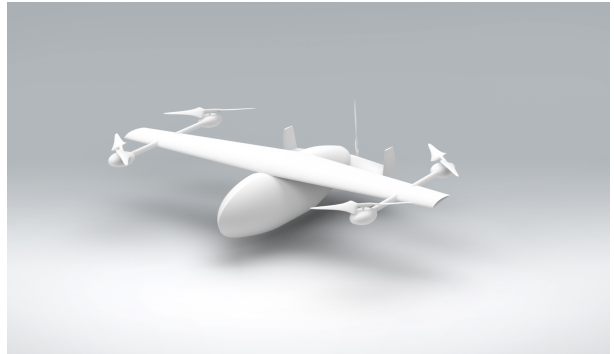


Figure 1.2: Render of baseline configuration.

Table 1.1: Key parameters for *Fregata*.

MOTW / kg	51.5	Cruise velocity / kn	113
Maximum payload weight / kg	9.0	Stall speed / kn	58
Range / km	70	T/W (VTOL)	1.20
Internal combustion total power / kW	8.0	T/W (cruise)	0.22
Electric propulsion total power / kW	6.5	W/S / N m ⁻²	831

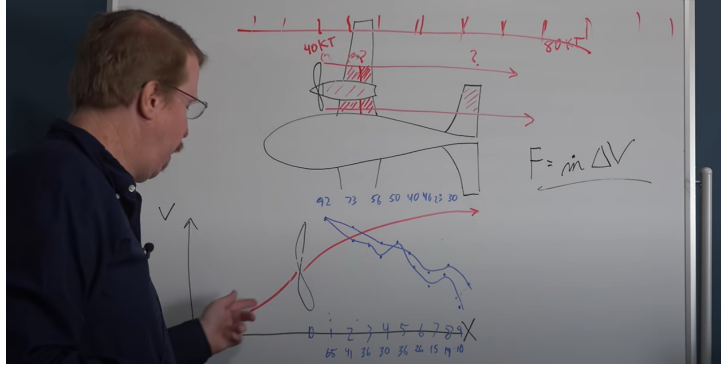


Figure 2.1: Presented data of the propwash velocity distribution from YouTube [11], with induced velocity in blue and theoretical distribution in red.

2.2 Flight Physics

The flight physics in X-Plane is driven by the blade element method (BEM). This involves initially partitioning the aircraft geometry into discrete elements, before the forces on each element are found using typical aeronautical engineering theory of two-dimensional lift, drag, and moments. These forces are calculated twice at each timestep and are then integrated to predict the motion of the aircraft [9]. This is a well-established methodology for finding forces on aerodynamic shapes, typically and historically used in rotor design, and builds up information on lifting surfaces from a close-to-first principles approach. This results in a very accurate determination of the forces present. Complementing this is a repository of additional models to capture effects that are missed by BEM, such as the three-dimensional effects, losses due to viscosity, and supersonic phenomena.

Notably, X-Plane is not open source, and so limited information is available regarding the exact models used in the software. Meyer has revealed some details on the additional models, where it was found that certain modelling features are implemented based on the creator’s own intuition, experience and experiments [10, 11]. Two pertinent examples of this are as follows:

- Fuselage forces are applied at a single point, which varies from 25% of the total fuselage length to 50% as the fuselage angle of incidence increases from 0° to 90° , as experimentally determined by Meyer using scale models. This assumption seems reasonable and lines up with flat plate theory, despite the test data and experimental setup not being available to assess. Notably, the magnitude of these forces are determined from literature related to long slender bodies, and it is uncertain as to whether models are included for bodies of lower fineness ratios.
- The induced velocity behind a propeller is modelled based on experimental tests using an anemometer behind an idling propeller on the ground. The resulting distribution is initially promising and agrees with engineering intuition, but the lack of repeated experiments at different power settings (or even the same power setting) and the fact that the aircraft was stationary could bring into doubt the model’s accuracy at different flight regimes, power settings, or even for different propeller designs. A screenshot from a YouTube video of the only presented data from the test can be seen in Figure 2.1, with flow on the left and right half of the fuselage plotted alongside the theoretical velocity distribution. Footage or results of equivalent tests in flight could not be found.

These are two examples that were seen to directly relate to the configuration of *Fregata*, and attention was made to any effects that may have arose due to modelling limitations such as propeller-wing interactions. Despite the large knowledge-base for quadcopter [12] and fixed wing aircraft [13] sizing and design, limited standard practises exist for vehicles with a hybrid configuration. Wing-propeller interactions for horizontal-axis propellers have been studied since antiquity, with Prandtl investigating the effect of propellers close to wings [14], but historically limited use of vertical axis propellers in conjunction with conventional lifting surfaces has produced a similarly limited body of literature. Nevertheless, aircraft such as the Bell Boeing V-22 have resulted in some literature outlining the effect of propeller downwash on wing performance in hover and forward flight [15, 16]. Earl [17] has carried out a vortex-lattice simulation investigating the effect of propeller vertical placement on the main wing, finding that a lower propeller placement was more favourable. The true effect of the propellers on the wing would be assessed

by looking at the transition performance and comparing it to predictions, as this is where these effects are most evident.

Despite these comments, widespread adoption and praise of X-Plane suggests that the modelling in the program does indeed hold in the wider domain of aviation, and so no steps were taken to adapt or improve any of the modelling carried out in the program.

2.3 Plane Maker and Airfoil Maker

X-Plane can be used with two proprietary programs to allow for design and implementation of user-defined aircraft. The first is Plane Maker, where the geometry of the vehicle is defined parametrically. This allows for the construction of a very detailed model of the aircraft, with detailed planform and twist information being used for lifting surfaces, and a point cloud used for the fuselage and other bodies. This meant that our aircraft geometry could be accurately and faithfully reflected in the program for the physics model to interpret.

The second program is Airfoil Maker, where aerofoils are defined to be used in the flight model. This results in accurate aerodynamic behaviour for the lifting surfaces and the propeller at different Reynolds numbers and flight conditions. It allows for a great deal of customisation, with the behaviour being parameterised for the lift, drag, and moment coefficients with angle of attack. Alongside good detail to model the aerofoil behaviour in positive and negative angles of attack, the behaviour of the post-stalled aerofoil is also modelled, with the distributions as seen in Figure 2.2. Meyer explains that the forces in this regime are similarly found using BEM, with an additional velocity perturbation given by a random distribution with an amplitude and frequency as determined by Meyer’s experience [10]. This was considered to be potentially inaccurate, and its accuracy for very small aircraft was uncertain. Nevertheless, the post-stall behaviour of the aircraft was not within the scope of this study and any results obtained in this regime were only used qualitatively.

3 Modelling and Implementation

3.1 Modelling in Plane Maker

Fregata was successfully modelled in Plane Maker. The conventional fuselage, wing, and empennage design were typical structures for the program and were able to be modelled effectively, with two separate vertical fins being used for the H-tail. Where appropriate, the maximum number of 10 elements were used as the discretisation to maximise modelling accuracy, while also allowing for the most accurate spanwise distribution for each control surface. It is reported that any finer discretisation results in a marginal increase in modelling accuracy with a greater increase in computational cost [10]. The modelling parameters for the main wing can be seen in Figure 3.1, with the discretised twist and aileron locations visible.

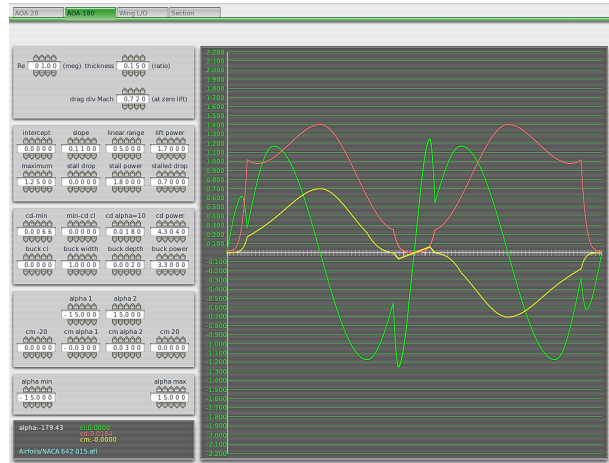


Figure 2.2: Entire parameterised lift, drag, and moment curve for NACA 642 – 015 between -180° and 180° angle of attack.

The VTOL structural configuration was less straightforward to model and required the use of miscellaneous bodies for the pylons and fairings. These are indeed considered in the physics model and should affect the dynamics of the aircraft, with X-Plane taking the drag coefficient of the pylons and applying the same physics model as the fuselage. This coefficient was set to be 0.014, from CFD simulations from Leung [18].

FOIL SPECS									
semi-length	0 0 1 1 0	(wing semi-length, root to tip, ALONG THE 25% CHORD, not span (m))	long arm	0 0 0 6 1	(m)	texture top			
root chord	0 0 0 4 4	(m)	sweep	0 0 0	(deg)	lat arm	0 0 0 0 0	(m)	left 0.7
tip chord	0 0 0 2 2	(m)	dihedral	0 0 0 0		vert arm	0 0 0 1 8	(m)	texture bot

ELEMENT SPECS (ROOT AT LEFT, TIP AT RIGHT)										☐ customize chords		
#	1 0	incidence	0 1 3	0 0 9	0 0 4	0 0 0	- 0 0 5	- 0 0 9	- 0 1 4	0 1 8	- 0 2 2	- 0 2 7
aileron 1	☐		☐	☐	☐	☒	☒	☒	☐	☐	☐	☐
aileron 2	☐		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
roll spoiler 1	☐		☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Figure 3.1: Plane maker parameters for main wing, with discretised twist implemented.

The propulsion system is somewhat native to Plane Maker, with options to define VTOL cyclic rotors and a throttle or collective differential for control inputs. To best reflect our aircraft, fixed propellers were used with geometry as designed by Frittoli [19] with an engine for each propeller, resulting in four electric engines and a single carburetor reciprocating engine to reflect the internal combustion engine (ICE). It was decided that the differential throttle inputs for control would be implemented in Simulink and not in Plane Maker, to allow for full freedom and transparency over their parameters. Alongside the propeller geometry, information on the power output and performance of the engines were taken to model the motors effectively, with efforts taken to match the predicted spool up time by changing the operating condition of the engine in Plane Maker. Notably, Plane Maker only asks for the parameters of a single engine, and so all the engines have the same performance in X-Plane. This is not the case for our aircraft, as the design maximum power output of the ICE is 6.375 kW higher than each of the electric ones. This discrepancy was resolved by reading the engine power from X-Plane and saturating the power output for the electric motors using Simulink, fixing them to their actual maximum power and not the power that Plane Maker has defined.

Weight and balance for the Plane Maker model was completed by obtaining the empty aircraft weight and centre of gravity position, before locating fuel tanks and payloads were appropriate. It was found that Plane Maker allows for droppable payloads to be designed and implemented in any models, and so a representative 9 kg payload was added to our aircraft.

The aerofoils were successfully parameterised in Airfoil Maker for the aerodynamic surfaces and the propellers. The final sized Plane Maker model can be seen in Figure 3.2, alongside the reference CAD geometry.

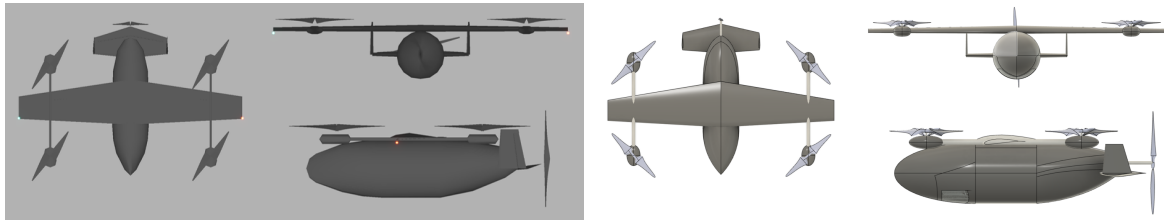


Figure 3.2: Plane Maker model of *Fregata* compared to working CAD model.

3.2 Communicating with X-Plane

X-Plane natively includes a User Datagram Protocol (UDP) for data communication to and from the software over the internet. This was initially used to collate data from each of the flight tests. Importantly, UDP can also send data to X-Plane such as control inputs, allowing controllers to be designed

and implemented directly into X-Plane, “closing-the-loop”.

Aside from UDP, X-Plane has an extensive software development kit (SDK) which allows development of plugins to change simulation parameters. This was also a feasible method of augmenting X-Plane and could reasonably implement a controller, but the lack of documentation for the SDK and the Simulation team’s unfamiliarity with the programming language C meant that it was thought to be unwise to try and develop plugins in the short time frame of the project.

Here, Simulink has been used to implement the UDP protocol in line with the majority of previous literature in the field [20, 21], as it further allowed for simple integration with the controllers designed by the Control subteam.

3.3 Simulink Implementation

This report only outlines the control implementation for the fixed-wing phase of the mission. For difficulties in implementing the VTOL control architecture, the reader is again directed to the report by Seyon. The model for fixed-wing flight had four main processes:

- UDP communication to and from X-Plane using the built in UDP blocks
- Controllers for lateral and longitudinal states for cruise.
- Generation of different reference signals for performance evaluation
- Processing results to produce a performance metric for each test

The UDP protocol was implemented successfully for this flight phase, and provided control surface deflections and the throttle setting for the pusher propeller. The structure of the complete Simulink model can be seen in Figure 3.3, with the two controllers highlighted.

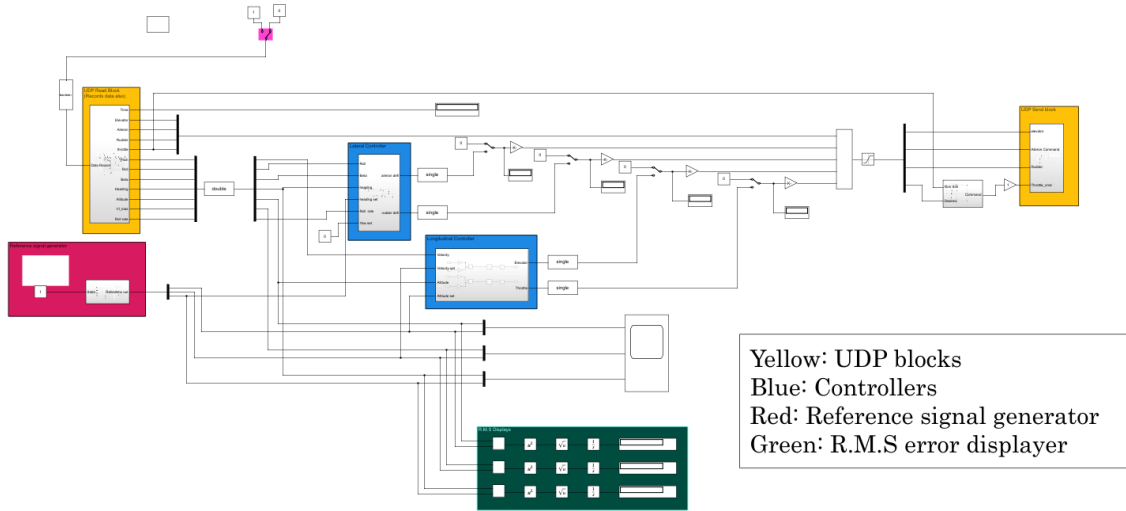


Figure 3.3: Complete Simulink model for closed loop testing with key block systems highlighted.

4 Flight Testing Scope

Despite longstanding and extensive literature on handling quality requirements for fixed-wing and rotary-wing aircraft, the rapid growth of the UAV sector has resulted in limited regulations for their behaviour in different regimes. Foster and Bowman [22] found that limits on time periods and damping ratios from a military specification (MIL-F-8785C) was unsuitable for small UAVs, due to the generally higher natural frequencies involved, and the authors developed their own standards using dynamic scaling. NASA is presently developing a full set of handling qualities for unmanned aerial systems, and the initial guidelines outlined by Klyde et al. [23] are used as a reference in this report. For *Fregata*, the relevant requirements are:

- all oscillations from trimmed flight must be stable for any airspeed within the flight envelope
- all oscillatory modes resulting from aerodynamic or control perturbations must have a damping ratio of at least 0.3
- no loss of stability should exist during transitions between fixed wing and VTOL flight modes

Some dynamic modes of the fixed wing aircraft - namely the phugoid, short period pitching oscillation (SPPO), and the Dutch roll - were investigated for both the open-loop and the closed-loop system. These were chosen as they were the most likely to be triggered in poor weather conditions, which is a performance requirement for our UAV, and would represent the most difficult conditions for the controllers and a remote pilot. Alongside these handling quality requirements, the tracking performance of the cruise controller will be identified using a series of reference trajectories with various altitude, heading, and velocity profiles. The chosen performance metric is the root-mean-square (R.M.S) error ϵ , given by

$$\epsilon = \int_0^T \sqrt{(r(t) - y(t))^2} dt. \quad (4.1)$$

No requirements were made into a maximum value of this function to meet the mission requirements, and it simply provided an easy-to-use metric to compare controller performance. A preliminary selection of gains for each controller was used initially, and a new set of tuned gains would be presented alongside the initially tuned ones if the performance was very poor.

Flight testing methods to excite the dynamic modes were taken from a US Navy Manual [24], designed for use in flight testing for military aircraft. It prescribes very clear methodologies for the triggering of each dynamic mode, and were found to often be suitable for our aircraft. Some changes were made due to the smaller scale of our UAV, such as using a single impulsive elevator input to trigger the phugoid due to difficulties in trimming the aircraft. Other considerations that came about from difficulties in testing are outlined in Section 5.

5 Flight Testing Results and Discussion

Flight testing was carried out over the course of a week on static flight simulators from Xesa Systems. This would be the first time that the aircraft was controlled by a pilot with a joystick and served to provide important feedback on the handling qualities of the vehicle. Table 5.1 outlines the key characteristics of the investigated oscillations, alongside predictions done by Chimonides [25] in the earlier stages of the design. The percentage difference for each value has been taken with reference to the predicted value.

5.1 Initial Impressions

The vehicle was very responsive in the fixed-wing configuration in the flight simulator, with good roll, pitch, and yaw authority. This sometimes resulted in a high pilot workload to correct any perturbations, and some example control inputs to maintain level flight can be found in Figure 5.1. Small pilot-induced oscillations can be seen between 9 and 12 s, reflecting difficulties in stabilising the aircraft. Using the Cooper-Harper scale [26] as a reference, the aircraft was given a rating of 5 for the open-loop fixed-wing mode, reflecting a general difficulty for control but still promising handling performance. For remote pilot operations of the UAV, it is suggested that the aircraft should not be flown without a controller in the periphery, such as an autopilot for cruise or a stability augmentation system for manoeuvring.

There was a general difficulty to trim the aircraft, especially as there are no trim tabs or an all moving tailplane. This was coupled with a noticeable torque on the aircraft due to the the pusher propeller, which results in a roll rate of roughly 14° s^{-1} with no control inputs. This initially made reaching steady level flight very difficult and hindered testing. This difficulty was overcome by introducing a virtual trim system in Simulink, where a user-defined elevator and aileron input could be superimposed to the joystick value, mimicking the effect of a trim wheel for pitch and roll. This was intuitive to use and resulted in a marked increase in stability and decrease in pilot workload, and was suggested as a design feature for the final aircraft.

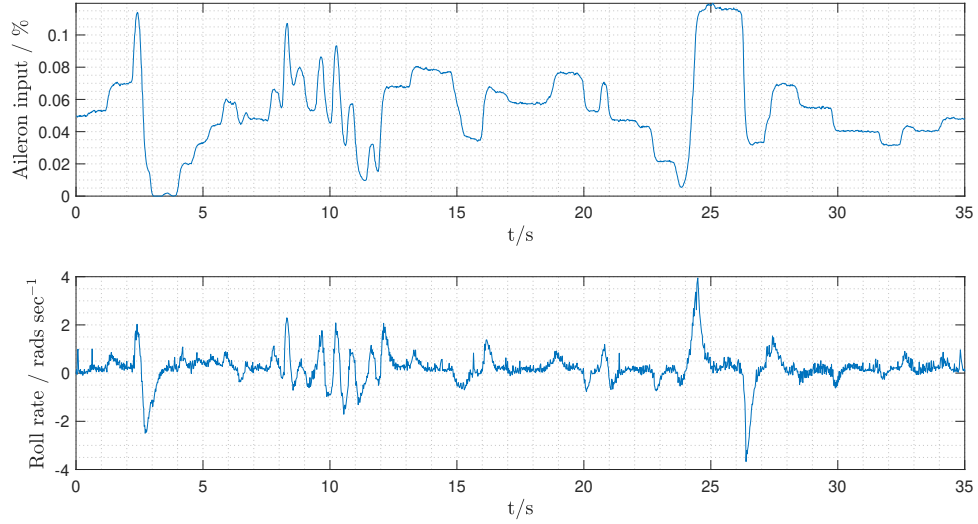


Figure 5.1: Sample control inputs for open loop system, with pilot-induced oscillations between 9 and 12 s.

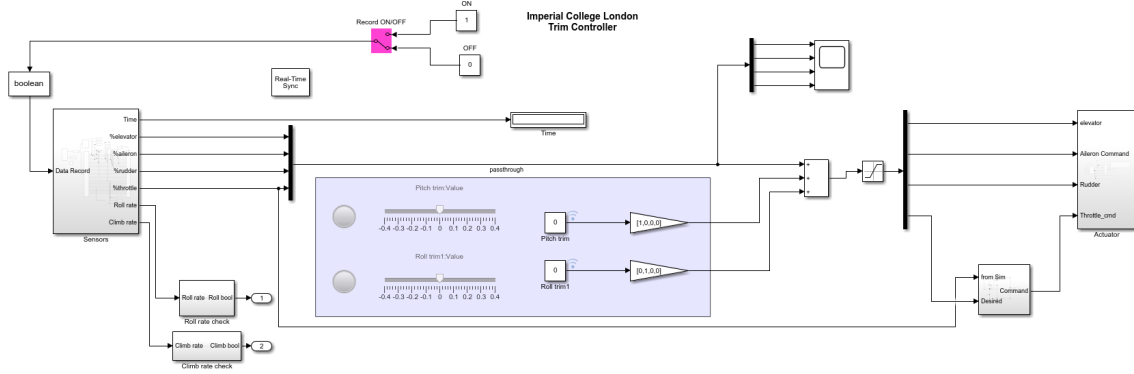


Figure 5.2: Virtual trim system used for testing, with trim sliders highlighted.

5.2 Phugoid Test

With the trim system as outlined previously, the aircraft showed reasonable ease to enter the phugoid motion following a sharp singlet with the elevator. It was found that the phugoid agreed well with predictions done by Chimonides, with a time period of 24.72 s and a slightly lower damping ratio of 0.04. Results from a sample phugoid can be found in Figure 5.3, where following the initial phugoid motion the aircraft was seen to try and deviate from the equilibrium altitude. It is thought that this is due to incorrect trim at the start of the motion, rather than deficiencies in the aerodynamic design, as the velocity behaviour is as expected.

If the initial elevator actuation was too aggressive, the aircraft was seen to deviate from the stable phugoid motion and stall within one and a half oscillations. An example of this is seen in Figure 5.4,

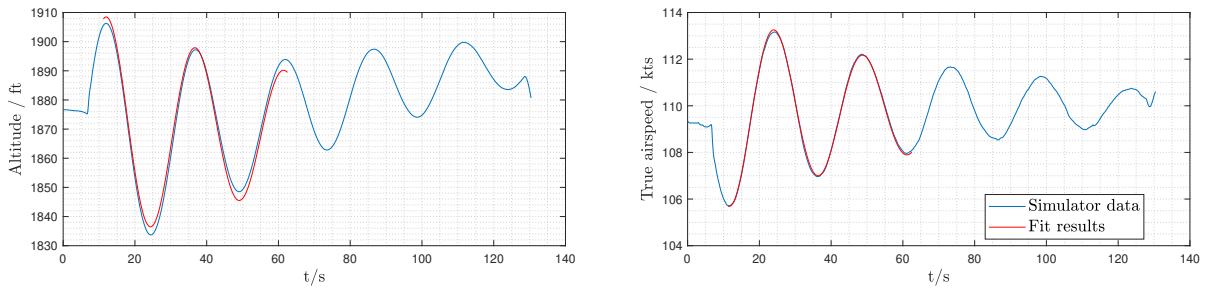
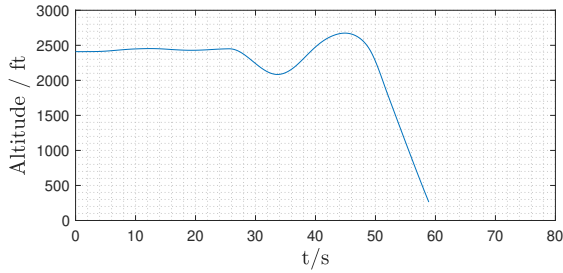
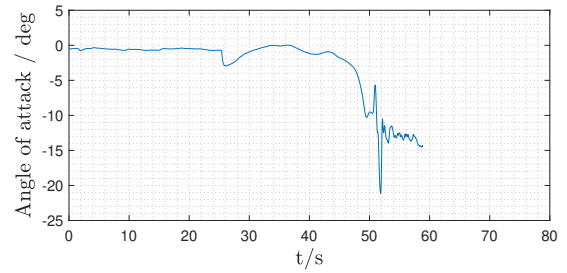


Figure 5.3: Sample phugoid plot showing slight instability after 80 s.

Table 5.1: Characteristics of predicted and simulation dynamic modes.

	Phugoid	SPPO	Dutch roll
Predicted eigenvalue	$-0.08 \pm 0.24i$	$-0.16 \pm 1.44i$	$-0.04 \pm 3.36i$
X-Plane eigenvalue	$-0.01 \pm 0.25i$	$-2.16 \pm 6.67i$	$-0.35 \pm 4.76i$
Eigenvalue difference* / %	1.1	383.9	42.0
Predicted frequency / rad s^{-1}	1.44	0.24	3.36
X-Plane frequency / rad s^{-1}	0.25	6.67	4.74
Frequency difference / %	82.6	2679.2	41.8
Predicted damping ratio	0.31	0.11	0.01
X-Plane damping ratio	0.04	0.31	0.07
Damping ratio difference / %	83.3	22.5	600
Predicted time period / s	26.56	4.34	1.87
X-Plane time period / s	24.74	0.94	1.32
Time period difference / %	6.8	78.3	29.4

*Eigenvalue difference presented is the percentage difference between absolute value of each eigenvalue.

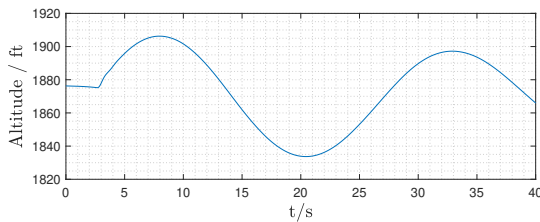
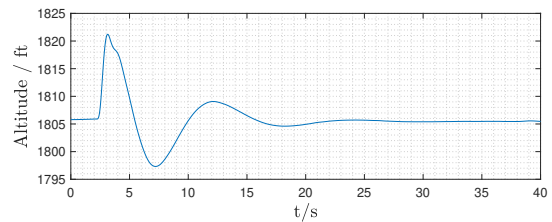
**(a)** Altitude variation of failed phugoid.**(b)** Angle of attack of poor phugoid, with stall at 49 s.**Figure 5.4:** Failed phugoid motion with stall following one oscillation.

where the aircraft stalled due to an excessive negative angle of attack of roughly -10° after one oscillation. It has been said that the post stall regime is potentially unreliable in X-Plane, and so these tests only served to show that the aircraft is sometimes able to exhibit dynamically unstable oscillations with too aggressive an input. A limit on the maximum elevator deflection of $\pm 15^\circ$ was advised to the Control subteam following these tests.

A pitch controller was designed by Yang [27] and was initially tuned using a Simulink model of the aircraft. Its effect on the phugoid was investigated. The closed-loop behaviour of the phugoid was stark, and minimal large period oscillations were seen by the aircraft. The initial variation in altitude following a singlet for the open loop and closed loop system can be seen in Figures 5.5a and 5.5b, with the motion disappearing within 25 seconds. This controller was not retuned to improve dynamic performance.

5.3 SPPO Test

It was found that the SPPO was very prone to stalling or entering an aggressive phugoid following the initial perturbation, and only small elevator doublets resulted in a distinct SPPO. From these, the behaviour was very damped, and was desirable. An example plot of an SPPO can be seen in Figure 5.6,

**(a)** Open-loop phugoid development.**(b)** Closed-loop phugoid development.**Figure 5.5:** Open-loop and closed-loop phugoid behaviour, with successful pitch control evident.

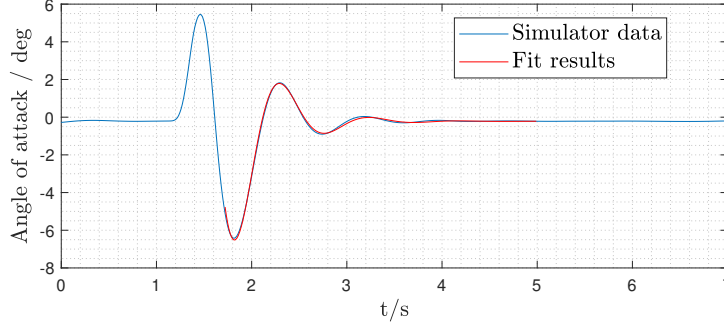
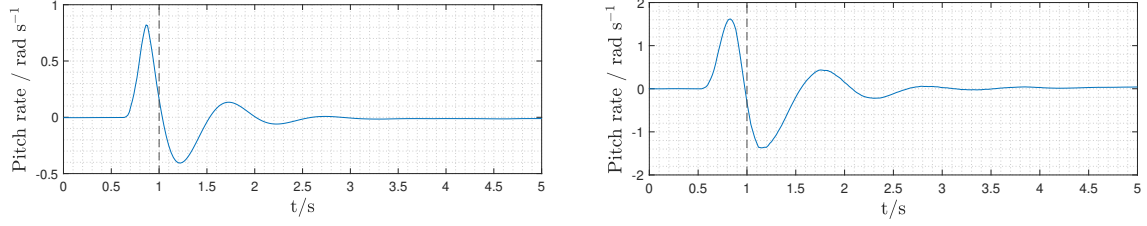


Figure 5.6: Sample SPPO plot with oscillatory fit results.



(a) Open-loop behaviour with oscillation starting at 1 s. **(b)** Closed-loop behaviour with oscillation starting at 1 s.

Figure 5.7: Open-loop and closed-loop SPPO behaviour showing little difference between the two oscillations.

with a time period of 0.94 s and a damping ratio of 0.31. It is worth noting that these differ largely from predictions, with the error in frequency being as large as 2679.2 %. This was found to be due to inaccurate predictions of the neutral point in previous analyses, and has been identified to be the reason why the damping of the SPPO is erroneously lower than that of the phugoid in predictions [25, 28]. Updating the neutral point from testing gives a damping of 0.400 for the SPPO and 0.240 for the phugoid, giving differences of 90.0% and 29.1% respectively.

The closed-loop behaviour of the SPPO was very similar to the open-loop behaviour, revealing that the controller was less successful at removing high-frequency perturbations. This is evident in Figures 5.7a and 5.7b, where both pitch rate oscillations have similar features. This was not a large weakness in the controller design, as the SPPO is already heavily damped, and would not hinder performance of the vehicle. Attempts to settle high-frequency perturbations often destabilised the aircraft, and is discussed further in in Section 5.5, so no attempts were made to improve the controller in this respect.

5.4 Dutch Roll

Despite difficulties to trim the aircraft in roll, good data was obtained for the Dutch roll. Most tests resulted in the development of a spiral in the absence of a corrective roll input due to incorrect roll trim. An example set of favourable data is seen in Figure 5.8, showing a short period of 1.32 s and a damping ratio of 0.07. Notably, this varies largely from values obtained in predictions, with the damping ratio being seven times higher in X-Plane. It was seen that the damping of our aircraft in both predictions and the simulation was in general lower than that of similarly sized aircraft [22]. This suggested that the aerodynamic configuration was somewhat lacking, and the damping could be improved by increasing the tail volume coefficient or moving the centre of gravity forward, as the tail is as aft as possible with imposed the fuselage constraints.

The initially presented roll and yaw controller was successful in reducing the Dutch roll oscillations, with the resulting motion being heavily damped rather than oscillatory, as compared in Figures 5.9a and 5.9b. This was promising and also indicated that the yaw damper did indeed reduce the effect of adverse yaw. The controller was so successful so as to hinder the triggering of the dutch roll, as can be seen in the oscillatory roll rates before 2 s in Figure 5.9b as the controller tried to resist the rudder inputs. This controller was also not retuned to improve dynamic behaviour.

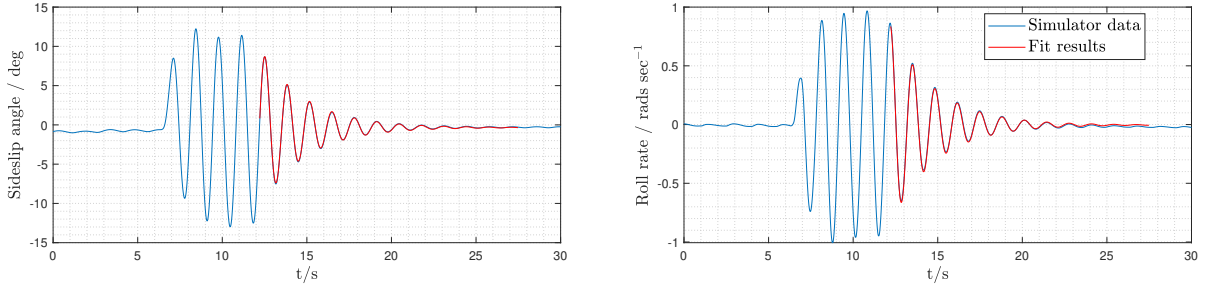


Figure 5.8: Sample Dutch roll plot with fit results.

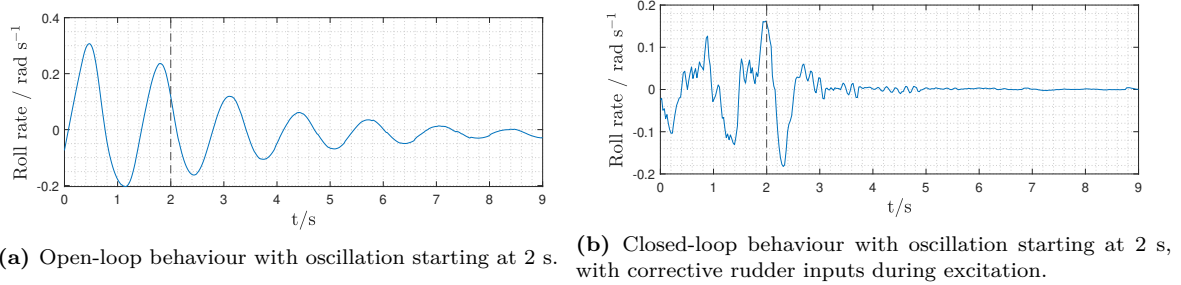


Figure 5.9: Open-loop and Closed-loop Dutch roll behaviour showing successful damping of Dutch roll.

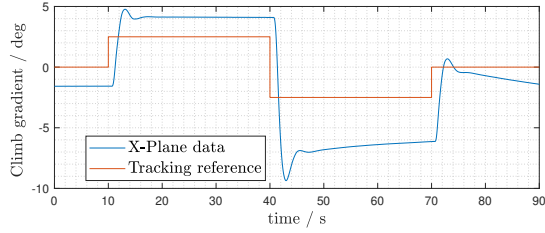
5.5 Fixed-Wing Controller Performance

Controllers were designed by Yang for the climb gradient, roll angle, yaw angle, and velocity, providing control inputs for the elevators, ailerons, rudder, and throttle respectively. These were all PID controllers, with proportional, integral, and derivative gains given by K_P , K_I , and K_D respectively. As well as their effect on the aircraft's dynamics as presented, the tracking accuracy of each controller was investigated. In line with NASA requirements, these tests were carried out at a range of airspeeds from 70 kn to 110 kts to investigate their performance at different flight phases. It was found that the lack of an integral gain in some controllers resulted in a noticeable steady-state error, and a brief retuning process was carried out. The initial and retuned gains are presented in Table 5.2.

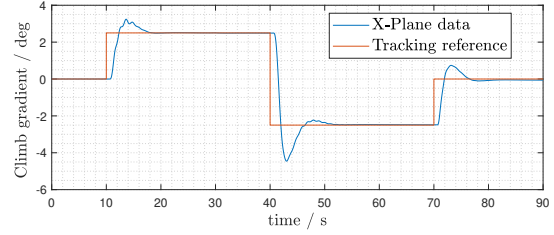
The pitch controller showed acceptable performance, which required minor tuning besides the addition of an integral gain. The improvement in performance with tuning can be seen in Figure 5.10, with the steady-state error disappearing entirely at cruise speed. The development of the final gains can be seen in Appendix A, with the development of oscillatory behaviour with the derivative gain evident. There was general difficulty in tuning the controller for robustness with respect to flight velocity, and the final controller displayed poor behaviour at speeds close to stall. An example of this is seen in Figure 5.11a, where the controller clearly struggled to maintain pitch in a 2° climb at 70 kn. This was seen to be due to the fact that the velocity very sharply decreased following the elevator command, resulting in an initial decrease in velocity of 9 kn as seen in Figure 5.11b. The controller consequently failed to trim the aircraft in the climb as multiple states were continually changing. Furthermore, the velocity controller was inadequate to overcome this and full throttle command only resulted in a slight improvement

Table 5.2: Controller gains before and after simulator testing.

Controller	K_P	K_I	K_D
Climb gradient initial	0.0025	0	0.005
Climb gradient retuned	0.004	0.001	0
Roll initial	0.02	0	0.003
Roll retuned	0.01	0.001	0.005
Yaw initial	0.05	0.001	0.05
Yaw retuned	0.05	0.00	0
Velocity initial	0.1	0.01	0
Velocity retuned	0.1	0.07	0.1

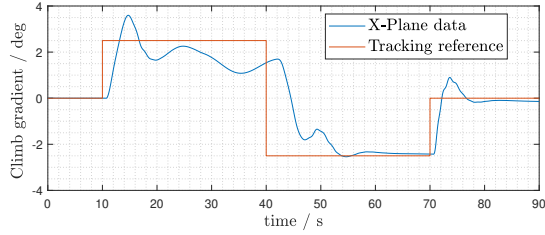


(a) Tracking performance for initially tuned pitch controller, with steady-state error evident. The controller was turned off at 75 s resulting in deviation.

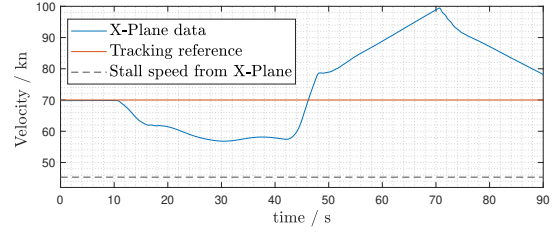


(b) Final pitch controller.

Figure 5.10: Tracking for final altitude controller with improved performance.



(a) Poor controller performance at 70 kn.



(b) Deviation from initial velocity during climb resulting in general climb difficulty.

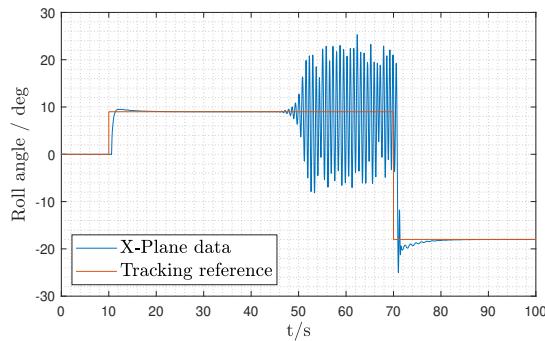
Figure 5.11: Controller performance at 70 kn, with velocity deviation shown with stall speed.

in the deceleration. Increasing the gains or increasing the maximum elevator deflection to counteract this resulted in aggressive inputs at high speeds, often resulting in unstable oscillations leading to stall. As such, the gains were tuned for best performance at the cruise condition. It was suggested that an adaptive controller could be designed, with variable gains and elevator saturation depending on the flight speed.

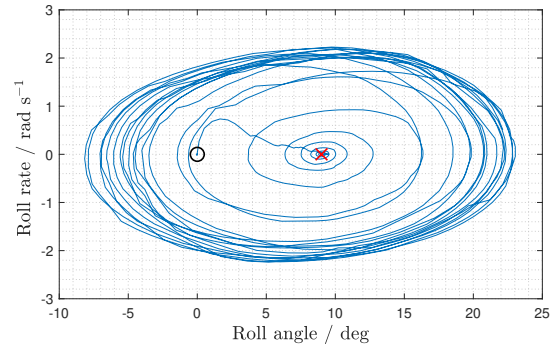
It is also important to note that the reference signal of an instantaneous step is extreme, and would not reflect any reference that *Fregata* would encounter during a mission. This was further motivation to conclude that the pitch controller was successful.

The velocity controller was successful and was only retuned to eliminate the steady-state error. Interestingly, it heavily affected the roll of the aircraft due to the high-frequency changes in propeller thrust and the resulting rolling torque. This resulted in variations in roll during velocity tracking tests, which were minimal and successfully corrected by the roll controller.

Interestingly, the roll controller initially displayed unstable behaviour when turning against the pro-



(a) Tracking performance for unstable roll behaviour.



(b) Phase portrait of unstable roll controller behaviour, with step starting at the circle marker and ending at the cross.

Figure 5.12: Unstable roll controller response to reference signal, with limit cycle phase portrait.

PELLER torque and was steady when turning with the propeller torque, with a left-turn being stable and a

right-turn repeatedly decaying into a stable limit cycle oscillation following no obvious perturbation, as seen in Figures 5.12a and 5.12b. This occurred at a range of flight speeds and reflects effect of the subtle asymmetry of our aircraft on the control system, resulting in a maximum roll rate of $\pm 2 \text{ rads s}^{-1}$ and roll angle of $\pm 15^\circ$ from the setpoint at 110 kn. This was clearly an undesirable phenomenon and was eliminated with a retune of the roll controller gains. This behaviour was successfully eliminated and did not reemerge at other flight speeds. A sample plot of the successful roll controller can be seen in Figure 5.13, and while a asymmetry in each direction still displayed, the final controller is stable always.

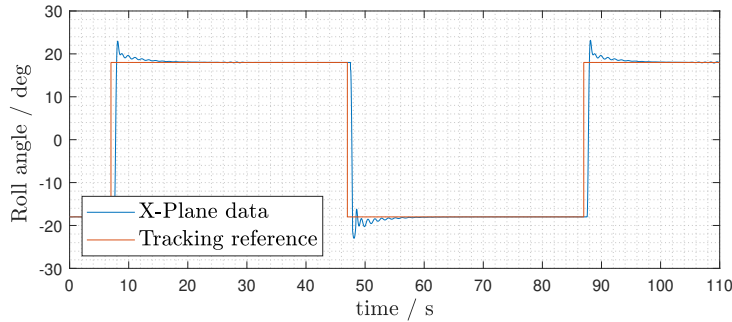


Figure 5.13: Successful roll control with unstable behaviour eliminated.

6 Conclusions

X-Plane 10 has been used to investigate the dynamic behaviour and the tracking performance of *Fregata* in the fixed-wing mode. The software was seen to satisfactorily model the physics of the vehicle and the proprietary modelling software were used successfully. The dynamic performance has been seen to differ from predictions done using state-space methods, and differences were seen to have mainly come about due to inaccuracies in predicting the aircraft neutral point and a small tail arm. These can be resolved with a larger tailplane and better modelling methods in earlier design. Aggressive pitch authority was seen due to the low static margin and should be resolved with, smaller elevators, a tailplane resize, and assistive systems for pilot-in-the-loop control.

The designed controllers were generally successful. The yaw and roll controllers successfully suppressed the Dutch roll and were retuned to eliminate an asymmetric instability in tracking behaviour. The velocity controller was similarly retuned to eliminate steady state errors and was otherwise successful. It was seen that the pitch controller was not satisfactorily robust following some retuning, and was less effective at low speeds. This was acceptable and suggestions have been made regarding adaptive controllers for different flight regimes.



References

- [1] Karamanou A, Drelioni GC, Papadimitriou D, Hahlikakis A. *Supporting Search and Rescue Operations with UAVs*. 5th International Conference on Civil Protection New Technology, Kozani. 2018.
- [2] Idrissi M, Salami M, Annaz F. *A Review of Quadrotor Unmanned Aerial Vehicles: Applications, Architectural Design and Control Algorithms*. Journal of Intelligent Robotic Systems, 2022. 104: 22.
- [3] Group Design Project. *Flying Liferaft: Design Specification*. 3rd year Group Design Project Executive Summary. Department of Aeronautics, Imperial College London. 2022.
- [4] Seyon G. *Simulation and Testing of Autonomous UAV for Emergency Aid Deployment at Sea*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.
- [5] Laminar Research. *X-Plane 10 Desktop Manual*. <https://www.x-plane.com/manuals/desktop/10/> [Accessed 11/06/2022]
- [6] Meyer A. *Austin Meyer - Developer & Filmmaker*. <https://austinmeyer.com/> [Accessed 07/06/2022]
- [7] Davoudi B, Duraisamy K. *A Hybrid Blade Element Momentum Model for Flight Simulation of Rotary Wing Unmanned Aerial Vehicles*. AIAA Aviation 2019 Forum, Texas. 17-21 June 2019.
- [8] Garcia R, Barnes L. *Multi-UAV Simulator Utilizing X-Plane*. Journal of Intelligent & Robotic Systems 57.1-4, 2009: 393–406.
- [9] Laminar Research. *How X-Plane Works*. <https://www.x-plane.com/desktop/how-x-plane-works/> [Accessed 15/06/2022]
- [10] Meyer A. Vertical Flight Society. *Austin Meyer How X Plane can be used for eVTOL Flight, presented by the eVTOL Flight Test Council*. Uploaded 3rd February 2021. <https://www.youtube.com/watch?v=DjXPqnmhMOA>
- [11] Meyer A. Michael Brown, XForcePC. *Testing Prop Wash on the Ground for X-Plane Flight Model*. Uploaded 15th March 2020. <https://www.youtube.com/watch?v=DjXPqnmhMOA>
- [12] Harkare O, Maan R. *Design and Control of a Quadcopter*. International Journal of Engineering and Technical Research, 2021. 10(5):257.
- [13] Gundlach J. *Designing Unmanned Aircraft Systems: A Comprehensive Approach*. Reston, Virginia: American Institute of Aeronautics and Astronautics, Inc., 2014.
- [14] Prandtl L. *Mutual Influence of Wings and Propeller*. First report of the Gottingen Aerodynamic Laboratory, Chapter 4, Section 6. 1921.
- [15] Lim J. *Fundamental Investigation of Proprotor and Wing Interactions in Tiltrotor Aircraft*. VFS 75th Annual Forum & Technology Display, Philadelphia, Pennsylvania. 2021.
- [16] Derby MR, Young LA. *Rotor/Wing Interactions in Hover*. NASA/TM-2002-211392, Ames Research Center. 2002.
- [17] Earl D. *Wing Planform Optimisation and Propeller Interaction Analysis*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.
- [18] Leung YK. *Initial Wing Planform Design, Nose design and Star CCM+ analysis on the Aerodynamic Performance of Fregata*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.
- [19] Frittoli J. *UAV Propeller Analysis and Actuator Sizing*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.

- [20] Bittar A, Figueiredo HV, Mendes AC. *Guidance Software-In-The-Loop simulation using X-Plane and Simulink for UAVs*. 2014 International Conference on Unmanned Aircraft Systems (ICUAS). IEEE, 2014. 993–1002.
- [21] Figueiredo HV, Saotome O. *Simulation Platform for Quadricopter: Using Matlab/Simulink and X-Plane*. 2012 Brazilian Robotics Symposium and Latin American Robotics Symposium. IEEE, 2012. 51–55.
- [22] Foster TM, Bowman WJ. *Dynamic Stability and Handling of Qualities of Unmanned-Aerial-Vehicles*. 43rd AIAA Aerospace Sciences Meeting and Exhibit. 2005.
- [23] Klyde DH, Schulze CP, Miller JP, Manriquez JA, Kotikalpudi A, Mithcell DG et al. *Defining Handling Qualities of Unmanned Aerial Systems: Phase II Final Report*. National Aeronautics and Space Administration, Report number NASA/CR-2020-220564, 2020.
- [24] US Naval Test Pilot School. *Fixed Wing Stability and Control*. Naval Air Warfare Center, USNTPS-FTM-No. 103. 1997.
- [25] Chimonides EC. *Open Loop Dynamic Stability*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.
- [26] Cooper GE, Harper RP. *Use of Pilot Rating in the Evaluation of Aircraft Handling Qualities*. National Aeronautics and Space Administration, NASA TN D-5153. 1969.
- [27] Yang B. *NAME*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.
- [28] Puaah YE. *Empennage Design and Static Stability Analysis*. 3rd year Group Design Project Report. Department of Aeronautics, Imperial College London. 2022.

A Retuning Process for Pitch Controller

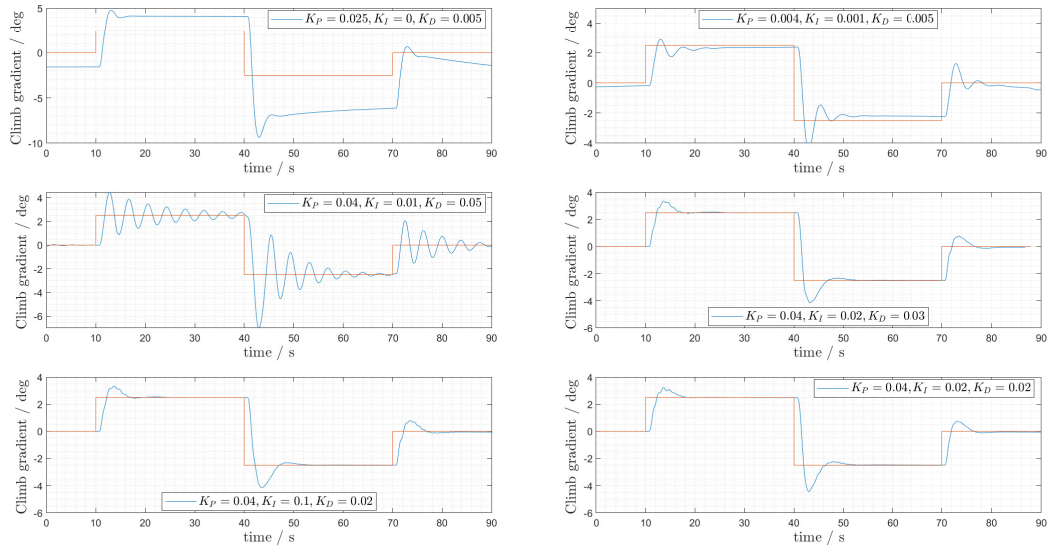


Figure A.1: Retuning process for pitch controller, with final controller at bottom right.